

# Marika Nishi

Robotics Application Engineer

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## EDUCATION

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| <b>University of Pennsylvania, School of Engineering &amp; Applied Science</b>   Philadelphia, PA, United States | May 2027      |
| Candidate for Master of Science in Engineering, <i>Majors: Robotics, Computer Science</i>                        | GPA: 3.81/4.0 |
| <b>University of Tokyo, Faculty of Engineering</b>   Bunkyo, Tokyo, Japan                                        | March 2023    |
| Bachelor of Engineering, <i>Major: Mechanical Engineering</i>                                                    | GPA: 3.64/4.0 |

## SKILLS & RELEVANT COURSEWORK

**Software Skills:** Python, PyTorch, C++, C, ROS, ROS2, Linux, Docker, Git, protobuf, SQL, Jupyter notebooks, AI, ML, EKF, sensor fusion, motion planning, autonomous driving, control, drones, perception, SLAM, simulation (Gazebo, Isaac Sim)

**Hardware Skills:** LiDAR, RTK-GNSS, GNSS, IMU, CAN bus, RGB cameras, depth sensors, motors, microcontroller

**Relevant Coursework:** Mechatronics; F1tenth Autonomous Racing Cars; Automotive Engineering; Deep Neural Networks; Deep Learning; Computer Vision; Dynamics and Control; Machine Perception; Machine Learning

**Publication:** Hori, K., Watanabe, K., Tanaka, T., **Nishi, M.**, & Ito, T. (2026). Visibility Evaluation Around Unsignalized Intersections with Occlusion From Arbitrary On-road Viewpoints Using Point Cloud Maps. *International Journal of Intelligent Transportation Systems Research*.

Watanabe, K., **Nishi, M.**, & Ito, T. (2025). Placement design of various roadside sensors for stochastic position prediction of traffic participants on community roads. *International Journal of Intelligent Transportation Systems Research*.

## EMPLOYMENT EXPERIENCE

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| Vijay Kumar Lab   <i>Research Assistant</i> , Philadelphia, PA, United States                                                                                                                                                                                                                                                                                                                                                                                                                             | May 2026 – Present           |
| <ul style="list-style-type: none"><li>Design and build VLA control system of drones with robotic arms to complete tasks like turning valves with IsaacSim</li></ul>                                                                                                                                                                                                                                                                                                                                       |                              |
| ISUZU   <i>Advanced Engineering - AI &amp; Autonomous Driving Intern</i> , Plymouth, MI, United States                                                                                                                                                                                                                                                                                                                                                                                                    | May 2026 – August 2026       |
| <ul style="list-style-type: none"><li>Investigated autonomous truck on-road test failures by tracing issues across a large-scale perception, planning, and control software architecture built with Bazel and protobuf, identifying root causes through code analysis and debugging</li><li>Used SQL to analyze autonomous driving on-road test data and quantitatively evaluate performance on dashboards</li></ul>                                                                                      |                              |
| Vijay Kumar Lab   <i>Research Intern</i> , Philadelphia, PA, United States                                                                                                                                                                                                                                                                                                                                                                                                                                | May 2025 – May 2026          |
| <ul style="list-style-type: none"><li>Implemented driving simulators that visualized data of event camera on F1tenth car using Gazebo, CARLA, and ROS</li><li>Collected datasets on driving simulator and trained vision transformer model that computed appropriate motion control from event camera images; improved data generation process efficiency by removing redundancy, cutting time by 70 %</li></ul>                                                                                          |                              |
| BOSCH   <i>Software Intern</i> , Tsuzuki, Kanagawa, Japan                                                                                                                                                                                                                                                                                                                                                                                                                                                 | August 2023 – September 2023 |
| <ul style="list-style-type: none"><li>Designed and built map to enhance autonomous driving by visualizing key information, such as routes and vehicle state using MATLAB; helped company's engineers understand signals sent from cars visually</li><li>Improved software that processed CAN signals from cars by fixing signal decoding; refined system output accuracy</li><li>Collaborated with colleagues from 10 countries and different majors; made outcomes reflecting diverse opinions</li></ul> |                              |
| DMG MORI   <i>Industrial Practices Intern, Additive Manufacturing R &amp; D</i> , Iga, Mie, Japan                                                                                                                                                                                                                                                                                                                                                                                                         | August 2021 – September 2021 |
| <ul style="list-style-type: none"><li>Analyzed performance tests of additive manufacturing machines by operating the machines and inspecting products</li><li>Investigated additive manufacturing methods by examining machine structures; proposed solutions to heat distortion</li></ul>                                                                                                                                                                                                                |                              |

## ROBOT AUTOMATION PROJECTS

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| F1tenth Autonomous Racing Cars   <i>Perception, Planning, Control, Autonomous Driving, Python, C++, ROS2</i>                                                                                                                                                                                                                                                                                                                                                                       | January 2025 – May 2025 |
| <ul style="list-style-type: none"><li>Developed a 1/10-scale autonomous racing car; implemented SLAM localization and supported integration and testing of motion planning and vehicle control systems using Python, C++, and ROS; won second place for fastest lap in race</li></ul>                                                                                                                                                                                              |                         |
| Research of Traffic Collision Prediction System   <i>Prototyping, Extended Kalman Filter, C++, ROS</i>                                                                                                                                                                                                                                                                                                                                                                             | April 2022 – March 2023 |
| <ul style="list-style-type: none"><li>Designed and built real-time collision prediction system among pedestrian, cyclist, and intelligent wheelchair</li><li>Wrote C++ ROS codes that detected pedestrians and cyclists based on LiDAR data</li><li>Conducted sensor fusion using Extended Kalman Filter; estimated pedestrians and cyclists' positions</li><li>Operated RTK-GPS, low-accuracy GPS, LiDAR, and IMU to collect and integrate data into system development</li></ul> |                         |